



Published in final edited form as:

*Am J Prev Med.* 2021 November ; 61(5 Suppl 1): S130–S142. doi:10.1016/j.amepre.2021.05.029.

## HIV Cluster and Outbreak Detection and Response: The Science and Experience

Alexandra M. Oster, MD<sup>1,2</sup>, Sheryl B. Lyss, MD, MPH<sup>1,2</sup>, R. Paul McClung, MD<sup>1,2</sup>, Meg Watson, MPH<sup>1</sup>, Nivedha Panneer, MPH<sup>1</sup>, Angela L. Hernandez, MD, MPH<sup>1</sup>, Kate Buchacz, PhD<sup>1</sup>, Susan E. Robilotto, DO<sup>3</sup>, Kathryn G. Curran, PhD<sup>1</sup>, Rashida Hassan, MSPH<sup>1</sup>, M. Cheryl Bañez Ocfemia, MPH<sup>1</sup>, Laurie Linley, MPH<sup>1</sup>, Stephen M. Perez, PhD, RN<sup>1,2</sup>, Stanley A. Phillip Jr., DHA, MHA<sup>1</sup>, Anne Marie France, PhD, MPH<sup>1,2</sup>

<sup>1</sup>Division of HIV Prevention, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention (NCHHSTP), Centers for Disease Control and Prevention, Atlanta, Georgia;

<sup>2</sup>U.S. Public Health Service, Atlanta, Georgia;

<sup>3</sup>Division of State HIV/AIDS Programs, HIV/AIDS Bureau, Health Resources and Services Administration, Rockville, Maryland

### Abstract

The Respond pillar of the Ending the HIV Epidemic in the U.S. initiative, which consists of activities also known as cluster and outbreak detection and response, offers a framework to guide tailored implementation of proven HIV prevention strategies where transmission is occurring most rapidly. Cluster and outbreak response involves understanding the networks in which rapid transmission is occurring; linking people in the network to essential services; and identifying and addressing gaps in programs and services such as testing, HIV and other medical care, pre-exposure prophylaxis, and syringe services programs. This article reviews the experience gained through 30 HIV cluster and outbreak responses in North America during 2000–2020 to describe approaches for implementing these core response strategies. Numerous jurisdictions that have implemented these response strategies have demonstrated success in improving outcomes related to HIV care and viral suppression, testing, use of prevention services, and reductions in transmission or new diagnoses. Efforts to address important gaps in service delivery revealed by cluster and outbreak detection and response can strengthen prevention efforts broadly through multidisciplinary, multisector collaboration. In this way, the Respond pillar embodies the collaborative, data-guided approach that is critical to the overall success of the Ending the HIV Epidemic in the U.S. initiative.

---

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Address correspondence to: Alexandra M. Oster, MD, Division of HIV Prevention, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention (NCHHSTP), Centers for Disease Control and Prevention, 1600 Clifton Road, Mail Stop US8-2, Atlanta GA 30329. [aoster@cdc.gov](mailto:aoster@cdc.gov).

#### SUPPLEMENT NOTE

This article is part of a supplement entitled The Evidence Base for Initial Intervention Strategies for Ending the HIV Epidemic in the U.S., which is sponsored by the U.S. Centers for Disease Control and Prevention (CDC), U.S. Department of Health and Human Services (HHS). The findings and conclusions in this article are those of the author(s) and do not necessarily represent the official position of CDC or HHS.

## INTRODUCTION

The Ending the HIV Epidemic in the U.S. initiative, which aims to reduce new HIV infections in the U.S. by 90% by 2030, includes 4 pillars: Diagnose, Treat, Prevent, and Respond.<sup>1</sup> The Respond pillar includes activities also known as cluster and outbreak detection and response (i.e., cluster detection and response [CDR]) and offers a framework to guide focused implementation of proven HIV prevention strategies from all pillars where transmission is occurring most rapidly.

CDR, a longstanding public health approach to reducing illness and death from infectious diseases,<sup>2–4</sup> involves the following:

1. identifying rapid transmission or changing transmission patterns;
2. reviewing available information and gathering additional information to understand the factors associated with increased transmission and opportunities for improved prevention; and
3. swiftly responding to these changes to decrease transmission by augmenting services and resources for the affected population.

The presence of an HIV cluster is a signal of potential increased HIV transmission within a geographic area, subpopulation, or social network and indicates gaps in prevention or care. Tailoring and focusing services to the relevant transmission network and addressing these gaps helps to improve the health of people with HIV and stop transmission to others.

Before the Centers for Disease Control and Prevention (CDC) implemented prospective, systematic analysis to detect clusters in 2016–2017, few health departments (HDs) conducted routine HIV cluster detection, and HIV cluster response occurred sporadically.<sup>5–11</sup> Beginning in 2018, CDC required all CDC-funded HDs to detect and respond to HIV clusters.<sup>12</sup> Although HDs are in varying phases of implementing this work, capacity for CDR has increased substantially in the U.S.<sup>13</sup> The Ending the HIV Epidemic initiative provides additional funding to HDs to strategically advance HIV prevention, including, for the Respond pillar, to implement innovative approaches to rapidly detect and respond to clusters and outbreaks.<sup>14</sup>

## HOW ARE CLUSTERS AND OUTBREAKS OF HIV DETECTED?

Cluster detection employs multiple methods. Sometimes, HIV clusters and outbreaks are identified when an increase in diagnoses or a change in the characteristics of those affected is noticed by healthcare providers, community members, or HD staff conducting partner services who identify, locate, and notify sex and needle-sharing partners of people with HIV and facilitate evaluation and care.<sup>11,15–19</sup> HDs and CDC also analyze surveillance data to systematically identify HIV clusters of increased diagnoses in a geographic area, sometimes called time–space clusters.<sup>8,20–34</sup> Finally, analysis of molecular data reported to HDs can identify clusters of highly similar HIV sequences, indicating rapid transmission.<sup>35–42</sup> State and local HDs detect molecular clusters using tools such as Secure HIV-TRACE (Secure

HIV TRANsmission Cluster Engine)<sup>43</sup> and identify time–space clusters using CDC-provided or locally developed analysis programs.

These approaches have varying strengths and weaknesses. Time–space analysis can provide an early signal in geographic areas or populations with relatively few HIV diagnoses at baseline, but an increase in diagnoses does not always represent increased transmission. Molecular analysis better identifies geographically dispersed clusters and, in areas with higher baseline HIV rates, can identify clusters before the number of infections is large enough to be identified in other ways but is dependent on linkage to care and sequence reporting. Partner services identifies social networks but is limited by clients' ability or willingness to name partners. All approaches rely on diagnosis (ideally timely diagnosis) of at least some portion of the cluster, and no approach will capture the full extent of the transmission network. Using multiple approaches maximizes early detection of clusters and outbreaks and the identification of the larger network of people affected by rapid transmission.<sup>16–18,23,34,45–48</sup>

## WHY IS RESPONDING TO CLUSTERS AND OUTBREAKS IMPORTANT?

HIV is transmitted through networks. CDR offers opportunities to identify and reach people in networks affected by rapid transmission, including those with undiagnosed HIV, those with HIV who might not be accessing HIV care and other services, and those who do not have HIV but might be at risk. Effective prevention and care strategies can be tailored to reach all who are affected.

Clusters and outbreaks have shown rapid and sometimes explosive transmission. Individual clusters and outbreaks (detected using either molecular or other approaches) have had transmission rates exceeding 30 times the national rate, and transmission rates in molecular clusters identified through routine analysis of surveillance data average 8–11 times the national rate.<sup>23,35,49–52</sup> Rapidly growing clusters contribute disproportionately to new infections,<sup>53</sup> and responding to these clusters may slow HIV evolution toward variants with higher transmissibility.<sup>54</sup>

Response efforts should be scaled to the size and characteristics of the cluster or outbreak and the needs of the transmission network. Although large outbreak responses are often highly visible,<sup>55</sup> CDR also includes intervening in the many clusters detected while still small,<sup>13</sup> which can most effectively limit transmission.<sup>17,23</sup> Therefore, CDR spans routine, day-to-day efforts to identify and prioritize clusters; initial response to small clusters; and expanded or escalated response to more concerning clusters and outbreaks (Figure 1).

## WHAT POPULATIONS ARE AFFECTED BY CLUSTERS AND OUTBREAKS?

Because transmission is not uniform even within priority populations, CDR goes beyond identifying subpopulations with the highest number of HIV diagnoses. Analysis by race/ethnicity and transmission category shows that groups experiencing rapid transmission do not necessarily mirror those with higher numbers of diagnoses and that these groups change over time and show substantial geographic variation.<sup>13</sup> Therefore, ongoing, routine analysis by HDs is necessary to identify the networks experiencing rapid transmission. By

identifying these networks that are not being effectively reached by existing HIV prevention services, such ongoing analysis presents a vital opportunity to address health equity.

In recent years, the largest responses have been to outbreaks among people who inject drugs (PWID).<sup>55,56</sup> Most were detected through nonmolecular approaches, such as time-space analysis, partner services, or identification by providers or community members, because these approaches are particularly effective at detecting clusters when the baseline number of cases in a population is relatively low.

However, sexual transmission is the predominant mode of HIV transmission in the U.S. and in molecular clusters of rapid transmission.<sup>13,35,57</sup> Although responses to clusters and outbreaks related to sexual transmission have occurred,<sup>6,7,9,10,19,24,28,32,34,36,37,58,59</sup> these responses often do not garner as much attention or resources as do outbreaks among PWID and have not frequently been described in the literature. Using diverse methods to identify clusters in a variety of populations is important to build support for addressing gaps in services affecting networks experiencing rapid transmission regardless of the mode of transmission.

This manuscript reviews the experience gained through 30 HIV cluster and outbreak responses during 2000–2020 (Appendix Table 1, available online) and describes approaches for implementing the 3 core CDR strategies<sup>14</sup>: (1) establishing partnerships, processes, data systems, and policies to facilitate robust, real-time CDR; (2) investigating and intervening in transmission networks; and (3) identifying and addressing gaps in programs and services revealed by CDR. Although outbreak responses have occurred in several other countries,<sup>60</sup> this review highlights CDR experiences in North America disseminated through manuscripts, conference presentations, or public communications (e.g., health alerts).

## **APPROACHES FOR STRATEGY 1: PARTNERSHIPS, PROCESSES, DATA SYSTEMS, AND POLICIES TO FACILITATE ROBUST, REAL-TIME CLUSTER DETECTION AND RESPONSE**

### **Internal Health Department Partnerships**

Given the multidisciplinary and multisectoral nature of CDR and the need for rapid decision making for a response, strong partnerships and active engagement of staff with diverse areas of expertise and leaders with authority to implement change across program areas are essential. For most jurisdictions, this involves establishing a standing committee or other processes to routinely review data to inform detection and response. For example, Maryland established a cluster response team, including leadership and staff in HIV surveillance, sexually transmitted disease (STD) prevention, and the integrated HIV prevention and care services program.<sup>41</sup> Maryland, Virginia, and the District of Columbia, which often have clusters that cross their boundaries, have implemented data sharing agreements to facilitate cross-jurisdiction cluster detection and also hold cross-jurisdiction standing meetings, promoting stronger collaboration during cluster response.<sup>61</sup> Philadelphia, through its Demonstrating Expanded Interventional Surveillance (DExIS) program, engages staff from multiple units in case reviews to determine service gaps and develop immediate and

long-term action plans for response activities.<sup>40</sup> At the federal level, an HIV Outbreak Coordination Unit has promoted proactive cross-unit collaboration.<sup>62</sup>

During escalated responses, enhanced coordination is useful. West Virginia; Scott County, Indiana; and Seattle/King County have implemented Incident Command Structures<sup>63</sup> during escalated responses.<sup>11,52,64</sup> A local response team established during a Philadelphia outbreak was strengthened by including non-HIV partners, such as viral hepatitis, STD, opioid, immunization, and public health preparedness programs.<sup>30</sup> Enhanced collaboration and communication between local and state HDs and among HDs in bordering states have also been useful in addressing an outbreak among PWID in Northern Kentucky and Ohio,<sup>65,66</sup> and when requested, assistance from CDC and other federal agencies has facilitated escalated response.<sup>11,16,24,36,52,67</sup>

### External Partnerships

Involving local community members and organizations in planning, implementing, and evaluating response activities can strengthen the delivered interventions. Community members, including people with or at risk of HIV, provide critical input and perspective on challenges and barriers and can help HDs design messaging and materials to avoid further stigmatizing affected communities and deliver services effectively.<sup>16,39,40,52,68,69</sup> Most HIV clusters occur in populations that are marginalized. Therefore, working with organizations trusted by affected communities has been particularly important in responses to clusters in diverse communities.<sup>16,19,52,70–73</sup>

Local organizations, including HIV services providers, community-based organizations, and syringe services programs (SSPs), may first recognize clusters<sup>16,17</sup> and can serve as implementation partners well positioned to carry out response activities. Partnerships with community organizations during responses have led to expanded HIV testing in homeless shelters and encampments,<sup>16,64</sup> emergency departments,<sup>52,64,67,72</sup> jails,<sup>19,64</sup> and community-based organizations.<sup>52,64</sup> Partnerships with Ryan White HIV/AIDS Program providers and case managers have helped to meet the needs of populations experiencing social and structural barriers to care.<sup>23,52,64,66</sup> HDs have collaborated with SSPs and mental-health providers to expand services for PWID.<sup>16,23,52,64,67</sup> Partnerships with academic institutions have helped to strengthen analytic capacity for CDR.<sup>23,34,46,51,64,74,75</sup> In response to a molecular cluster in San Antonio, providers, community members, and HD staff established the End Stigma, End HIV Alliance, whose efforts led to a quadrupling of HIV testing, accelerated achievement of the goal of 7-day linkage to HIV care, and activities to reduce stigma.<sup>72</sup>

Outbreak response planning and training exercises have identified gaps and needs in these partnerships and have facilitated community outreach before an outbreak occurred.<sup>19,76,77</sup> Engaging community organizations before a cluster or outbreak occurs speeds the ability to respond, should one arise, and can lead to interventions that prevent future outbreaks.<sup>16,19,64,78</sup>

## **Expediting Availability of High-Quality and Timely Surveillance Data for Real-Time Decision Making**

The success of CDR depends on the availability of high-quality and timely surveillance data. In the U.S., commercial, public, and private laboratories report the results of tests performed for HIV diagnosis and care, including HIV sequences generated through drug resistance testing, to HDs. Increasing the rapidity of laboratory reporting and automation of HD data processing practices can improve the availability of timely data and speed detection of and response to clusters, increasing opportunities to avert additional cases.<sup>37,52,64,79</sup>

## **Analyzing, Integrating, Visualizing, and Sharing Data From Diverse Sources for Cluster Detection and Response**

Successful CDR requires analyzing and integrating data from multiple sources, including HIV, STD, and viral hepatitis surveillance; partner services; and care systems, while maintaining appropriate data protections. Because these data often exist in siloed systems, data integration can be a challenge for response.<sup>52,67,80</sup> Some HDs have developed innovative approaches to data integration and used new tools, such as MicrobeTrace, for visualization.<sup>19,23,41,44,75</sup> HDs have incorporated data sources beyond case surveillance and partner services data; a Philadelphia outbreak response included HIV behavioral surveillance, medical examiner, shelter and homeless outreach, and SSP data, providing information about social and environmental determinants.<sup>30</sup> Regional data sharing through the use of data sharing agreements and collaboration through the engagement of staff and leadership in regular meetings has improved services, care engagement, and viral suppression in cluster and outbreak responses in the District of Columbia and Cincinnati metropolitan areas.<sup>61,66</sup>

## **Creating and Maintaining Flexible Funding Mechanisms Capable of Supporting Response Efforts**

Successful CDR requires flexible approaches to allocating funding and resources. HDs have had success in leveraging diverse funding resources, including HIV surveillance and prevention funding; opioid, overdose, and injury prevention funding; Ryan White HIV/AIDS Program funding; public health preparedness funding; and state funds.<sup>16,41,52</sup> To improve the ability of programs to pivot and adjust priorities in response to evolving situations, CDC is supporting HDs to (1) proactively design funding agreements to allow their funded entities to shift resources to address clusters and outbreaks and (2) identify resources from diverse sources that can be made available in case of a cluster or outbreak.<sup>52,81</sup>

## **APPROACHES FOR STRATEGY 2: INVESTIGATING AND INTERVENING IN NETWORKS**

### **Understanding Networks**

Understanding HIV transmission networks is essential for identifying intervention opportunities. A detected cluster indicates a larger underlying network, which includes people with diagnosed and undiagnosed HIV and people who do not have HIV.<sup>36,82,83</sup> A variety of approaches, including review of existing HD data, HIV partner services

expansion, peer-recruitment strategies,<sup>84,85</sup> and qualitative and quantitative data collection, can elucidate the extent and characteristics of a network, social or structural barriers, challenges to accessing services, and opportunities for public health interventions. Table 1 provides examples of several approaches to understanding networks.<sup>58,59,86–90</sup>

### **Providing Support for Linkage to and Retention in Critical Services**

As factors facilitating HIV transmission, including challenges affecting people in HIV transmission networks, are revealed, this information should be used to strengthen linkage to critical services, including testing and subsequent retesting for HIV, hepatitis B, hepatitis C, and STDs; pre-exposure prophylaxis; SSPs; HIV medical care; and other essential support services (e.g., housing, social services). Incident infections in rapidly growing clusters originate equally from diagnosed but unsuppressed cases and undiagnosed infections, highlighting the importance of promoting retention in care, medication adherence, and viral suppression in addition to partner services and other case-finding activities.<sup>83</sup> Prioritizing people in clusters for linkage to care and treatment interventions can reduce onward HIV transmission,<sup>91</sup> and reaching all members of a cluster with these services is more effective than prioritizing certain people within a cluster.<sup>74</sup>

Multiple HIV responses have successfully linked people to needed services. After enhanced public health follow-up for a molecular cluster in British Columbia, the median viral load of cluster members decreased from 38,109 copies/mL to undetectable.<sup>37</sup> During a response, West Virginia implemented same-day HIV care appointments, enhanced data to care, and care coordination, improving viral suppression rates.<sup>52</sup> Viral suppression also improved after a response in North Carolina that included linkage to care<sup>23</sup> and a response in Miami, which included wraparound services, intensive linkage, and patient navigation.<sup>17</sup> Initiation of a new HIV care site in Indiana led to a viral suppression rate of 75%.<sup>71</sup> These responses also linked people to services such as hepatitis A vaccination, hepatitis B and C treatment, and medication for opioid use disorder.

## **APPROACHES FOR STRATEGY 3: IDENTIFYING AND ADDRESSING GAPS IN PROGRAMS AND SERVICES REVEALED BY CLUSTER DETECTION AND RESPONSE**

Amid a response, gaps in programs and services must be identified and addressed swiftly to limit transmission. Outbreaks have occurred even in areas that were believed to have strong prevention programs,<sup>15,16,18,30</sup> underscoring the opportunity for cluster response to inform changes to prevention programming and support efforts to pursue continuous quality improvement.

Gaps might be identified in the delivery of or access to testing, HIV and other medical care, pre-exposure prophylaxis, SSPs, or other services. Programmatic response efforts may include strengthening these core HIV services and redoubling efforts to address social and structural gaps relating to homelessness, poverty, or stigma, as exemplified by the responses discussed in this article. Addressing gaps in ways that are lasting is important, and such improvements, whether made during or after cluster response, are important for preventing

future clusters and outbreaks. Table 2 shows key examples of various interventions to address gaps in programs and services.<sup>92–95</sup>

Identifying and addressing gaps in services is not only relevant for large outbreaks. The DExIS program in Philadelphia provides a model for continually and prospectively assessing barriers faced by sentinel cases, including people in small clusters, and using that information to modify long-term policy strategies and funding aims.<sup>40</sup>

In addition to improving services, focused public health messaging is an important component of response. Numerous responses have also included important communications such as clinical advisories, dear colleague letters, health alerts, social media campaigns, and print materials intended for community members and providers that addressed important topics in HIV prevention, testing, and care and dispelled misconceptions or myths.<sup>16,18,21,24–26,29–32,40,52,64,96,97</sup>

## CLUSTER AND OUTBREAK RESPONSE OUTCOMES

Response interventions occur at individual, network, and programmatic levels. Given the complex, multilayered nature of response interventions and limitations in determining what outcomes would have been observed if no focused response activities had occurred, assessing the impacts of response interventions can be difficult. However, multiple responses have seen changes in diagnoses, transmission, care outcomes, and other key outcomes after implementing response efforts.

For example, in the Indiana outbreak, after the state public health emergency was declared, transmission fell below the epidemic threshold, suggesting that improving awareness of the outbreak, establishing an SSP, and deploying additional care services reduced transmission.<sup>49</sup> Quantifiable benefits were also seen in opioid use disorder treatment, hepatitis A virus vaccination, hepatitis C virus treatment, and endocarditis prevention.<sup>71</sup> Likewise, response efforts in Northeastern Massachusetts, Seattle, and West Virginia were followed by substantial decreases in new HIV diagnoses,<sup>16,52,64</sup> and the lack of new linked cases after a response in North Carolina suggested that response efforts ultimately prevented further transmission.<sup>23</sup> Numerous responses have also documented improvements in important intermediate outcomes, including HIV testing, HIV diagnosis, pre-exposure prophylaxis uptake, SSP utilization, linkage to care, retention in care, and viral suppression.<sup>17,18,21,23,30,37,52,64,71,72,98,99</sup>

## ADDITIONAL CONSIDERATIONS

Although CDR efforts have shown real, proven public health benefits, it is important to also consider any possible unintended consequences.<sup>100</sup> Some community members and HIV advocacy organizations have expressed concerns about the possible use of molecular data as evidence in legal cases against people with HIV. Many states have HIV criminalization statutes or use general statutes (e.g., reckless endangerment) to prosecute people for HIV exposure or transmission. These statutes and actions are not based on science and run counter to public health efforts.<sup>101</sup> Most states have strong protections and policies against

releasing public health data as evidence in legal proceedings, but specific protections vary widely.<sup>102</sup> CDC does not support the use of public health data for nonpublic health purposes.

## RESEARCH GAPS

Further research is needed to strengthen the strategies for CDR; research is particularly important to expand the understanding of effective response strategies and tools for implementing these strategies (Table 3). Implementation science research can elucidate how best to implement testing, treatment, and prevention interventions during a response and how to evaluate outcomes. Social science research can inform best practices for engaging community members, organizations, and other stakeholders in cluster and outbreak preparedness, planning, and response, including addressing HIV stigma, structural barriers, and social determinants of health. Refining methods for predicting and detecting clusters early can aid early response efforts. Across all of these domains, research can help to bolster the understanding of how to optimize CDR for clusters related to sexual transmission.

## CONCLUSIONS

The use of CDR can identify rapid transmission so that proven, effective strategies can be implemented in affected populations or networks. Efforts to address important gaps in service delivery revealed by CDR can strengthen prevention efforts broadly through multidisciplinary, multisector collaboration. In this way, the Respond pillar embodies the collaborative, data-guided approach critical to the overall success of the Ending the HIV Epidemic in the U.S. initiative.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

## ACKNOWLEDGMENTS

The authors acknowledge staff from the Centers for Disease Control and Prevention (CDC), state and local health departments, and other agencies and organizations who have contributed to cluster detection and response efforts, including state and local HIV surveillance and prevention staff, members of CDC's HIV Outbreak Coordination Unit, CDC surveillance and prevention project officers, and staff involved in special projects related to cluster detection and response.

The findings and conclusions in this manuscript are those of the authors and do not necessarily represent the official position of the Health Resources and Services Administration.

No financial disclosures were reported by the authors of this paper.

## REFERENCES

1. Ending the HIV epidemic. [HIV.gov](https://www.hiv.gov/federal-response/ending-the-hiv-epidemic/overview), HHS. <https://www.hiv.gov/federal-response/ending-the-hiv-epidemic/overview>. Updated June 2, 2021. Accessed June 24, 2020.
2. Principles of epidemiology in public health practice, third edition: an introduction to applied epidemiology and biostatistics. Centers for Disease Control and Prevention, HHS; May 18, 2012. <https://www.cdc.gov/csels/dsepd/ss1978/index.html>. Accessed August 10, 2021.

3. Mindra G, Wortham JM, Haddad MB, Powell KM. Tuberculosis outbreaks in the United States, 2009–2015. *Public Health Rep.* 2017;132 (2):157–163. 10.1177/0033354916688270. [PubMed: 28147211]
4. Scharff RL, Besser J, Sharp DJ, Jones TF, Peter GS, Hedberg CW. An economic evaluation of PulseNet: a network for foodborne disease surveillance. *Am J Prev Med.* 2016;50(5):S66–S73 suppl 1. 10.1016/j.amepre.2015.09.018. [PubMed: 26993535]
5. Centers for Disease Control and Prevention (CDC). HIV transmission among black college student and non-student men who have sex with men—North Carolina, 2003. *MMWR Morb Mortal Wkly Rep.* 2004;53(32):731–734. <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm5332a1.htm>. [PubMed: 15318156]
6. Centers for Disease Control and Prevention (CDC). HIV transmission in the adult film industry — Los Angeles, California, 2004. *MMWR Morb Mortal Wkly Rep.* 2005;54(37):923–926. <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm5437a3.htm>. [PubMed: 16177683]
7. Buskin SE, Ellis GM, Pepper GG, et al. Transmission cluster of multiclass highly drug-resistant HIV-1 among 9 men who have sex with men in Seattle/King County, WA, 2005–2007. *J Acquir Immune Defic Syndr.* 2008;49(2):205–211. 10.1097/QAI.0-b013e318185727e. [PubMed: 18769347]
8. Centers for Disease Control and Prevention (CDC). HIV infection among young black men who have sex with men—Jackson, Mississippi, 2006–2008. *MMWR Morb Mortal Wkly Rep.* 2009;58(4):77–81. <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm5804a2.htm>. [PubMed: 19194368]
9. Nett RJ, Bartschi JL, Ellis GM, et al. Two clusters of HIV-1 infection, rural Idaho, USA, 2008. *Emerg Infect Dis.* 2010;16(11):1807–1809. 10.3201/eid1611.100857. [PubMed: 21029556]
10. Centers for Disease Control and Prevention (CDC). Increase in newly diagnosed HIV infections among young Black men who have sex with men—Milwaukee County, Wisconsin, 1999–2008. *MMWR Morb Mortal Wkly Rep.* 2011;60(04):99–102. <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6004a3.htm>. Accessed Aug 10, 2021. [PubMed: 21293324]
11. Peters PJ, Pontones P, Hoover KW, et al. HIV infection linked to injection use of oxycodone in Indiana, 2014–2015. *N Engl J Med.* 2016;375(3):229–239. 10.1056/NEJMoa1515195. [PubMed: 27468059]
12. Funding opportunity announcement (FOA) PS18–1802: integrated human immunodeficiency virus (HIV) surveillance and prevention programs for health departments. Centers for Disease Control and Prevention; April 11, 2019. <https://www.cdc.gov/hiv/funding/announcements/ps18-1802/index.html>.
13. Oster AM, Panneer N, Lyss SB, et al. Increasing capacity to detect clusters of rapid HIV transmission in varied populations—United States. *Viruses.* 2021;13(4):577. 10.3390/v13040577. [PubMed: 33808053]
14. Funding opportunity announcement PS20–2010: integrated HIV programs for Health Departments to Support Ending the HIV epidemic in the United States. Centers for Disease Control and Prevention; April 20, 2020. <https://www.cdc.gov/hiv/funding/announcements/ps20-2010/index.html>.
15. Cranston K, Alpre C, John B, et al. Notes from the field: HIV diagnoses among persons who inject drugs - Northeastern Massachusetts, 2015–2018. *MMWR Morb Mortal Wkly Rep.* 2019;68(10):253–254. 10.15585/mmwr.mm6810a6. [PubMed: 30870405]
16. Alpre C, Dawson EL, John B, et al. Opioid use fueling HIV transmission in an urban setting: an outbreak of HIV infection among people who inject drugs—Massachusetts, 2015–2018. *Am J Public Health.* 2020;110(1):37–44. 10.2105/AJPH.2019.305366. [PubMed: 31725317]
17. Tookes H, Bartholomew TS, Geary S, et al. Rapid identification and investigation of an HIV risk network among people who inject drugs—Miami, FL, 2018. *AIDS Behav.* 2020;24(1):246–256. 10.1007/s10461-019-02680-9. [PubMed: 31555932]
18. Golden MR, Lechtenberg R, Glick SN, et al. Outbreak of human immunodeficiency virus infection among heterosexual persons who are living homeless and inject drugs - Seattle, Washington, 2018. *MMWR Morb Mortal Wkly Rep.* 2019;68(15):344–349. 10.15585/mmwr.mm6815a2. [PubMed: 30998671]

19. Sizemore L, Fill MM, Mathieson SA, et al. Using an established outbreak response plan and molecular epidemiology methods in an HIV transmission cluster investigation, Tennessee, January-June 2017. *Public Health Rep.* 2020;135(3):329–333. 10.1177/0033354920915445. [PubMed: 32228123]
20. Fitzmaurice AG, Linley L, Zhang C, Watson M, France AM, Oster AM. Novel method for rapid detection of spatiotemporal HIV clusters potentially warranting intervention. *Emerg Infect Dis.* 2019;25 (5):988–991. 10.3201/eid2505.180776. [PubMed: 31002076]
21. Atkins A, McClung RP, Kilkenny M, et al. Notes from the field: outbreak of human immunodeficiency virus infection among persons who inject drugs - Cabell County, West Virginia, 2018–2019. *MMWR Morb Mortal Wkly Rep.* 2020;69(16):499–500. 10.15585/mmwr.mm6916a2. [PubMed: 32324723]
22. Furukawa NW, Blau EF, Reau Z, et al. Missed opportunities for human immunodeficiency virus (HIV) testing during injection drug use-related healthcare encounters among a cohort of persons who inject drugs with HIV diagnosed during an outbreak-Cincinnati/Northern Kentucky, 2017–2018. *Clin Infect Dis.* 2021;72(11):1961–1967. 10.1093/cid/ciaa507. [PubMed: 32748940]
23. Samoff E, Mobley V, Hudgins M, et al. HIV outbreak control with effective access to care and harm reduction in North Carolina, 2017–2018. *Am J Public Health.* 2020;110(3):394–400. 10.2105/AJPH.2019.305490. [PubMed: 31944835]
24. Evans ME, Labuda SM, Hogan V, et al. Notes from the field: HIV infection investigation in a rural area - West Virginia, 2017. *MMWR Morb Mortal Wkly Rep.* 2018;67(8):257–258. 10.15585/mmwr.mm6708a6. [PubMed: 29494569]
25. Minnesota Department of Health. Health advisory: HIV outbreak in persons who inject drugs (PWID). Saint Paul, MN: Minnesota Department of Health; February 6, 2020. <https://www.health.state.mn.us/communities/ep/han/2020/feb3hiv.pdf>. Published Accessed January 21, 2021.
26. Health officials alert public to increase in HIV infections among people who use drugs. Multnomah County <https://multco.us/multnomah-county/news/health-officials-alert-public-increase-hiv-infections-among-people-who-use>. Accessed August 10, 2021.
27. Alaska Department of Health and Human Services. HIV update – Alaska. [http://www.epi.alaska.gov/bulletins/docs/b2019\\_06.pdf](http://www.epi.alaska.gov/bulletins/docs/b2019_06.pdf).
28. New York State Department of Health. Health advisory: increased number of HIV and STI diagnoses in Orange County. Albany, NY: New York State Department of Health, September 10, 2018 [https://www.health.ny.gov/diseases/aids/providers/health\\_advisories/docs/september\\_2018.pdf](https://www.health.ny.gov/diseases/aids/providers/health_advisories/docs/september_2018.pdf). Published Accessed January 21, 2021.
29. Tri-County Health Department. Advisory - Increase in HIV in Colorado, especially among females and people who inject drugs. Brighton, CO: Tri-County Health Department. <https://www.tchd.org/DocumentCenter/View/6382/HAN-Advisory—Increase-in-HIV-in-Colorado-especially-among-females-and-people-who-inject-drugs—August-12-2019>. Published August 12, 2019. Accessed January 21, 2021.
30. Kim MM, Conyngham SC, Smith C, et al. Understanding the intersection of behavioral risk and social determinants of health and the impact on an outbreak of human immunodeficiency virus among persons who inject drugs in Philadelphia. *J Infect Dis.* 2020;222: S250–S258 suppl 5. 10.1093/infdis/jiaa128. [PubMed: 32877552]
31. Brown CM, Jaeger JL. Increase in newly diagnosed HIV infections among persons who inject drugs in Boston. Boston, MA: Massachusetts Department of Public Health and Boston Public Health Commission; January 25, 2019. [https://www.bphc.org/whatwedo/infectious-diseases/Documents/Joint\\_HIV\\_in\\_PWID\\_advisory\\_012519%20\(1\).pdf](https://www.bphc.org/whatwedo/infectious-diseases/Documents/Joint_HIV_in_PWID_advisory_012519%20(1).pdf). Published Accessed January 21, 2021.
32. Levine R Recent increases in reported HIV and gonorrhea in Beaver County. Harrisburg, PA: Pennsylvania Department of Health; March 2, 2018. [https://www.health.pa.gov/topics/documents/han/2018-pahan-400-03-02-adv\\_hiv%20sak.pdf](https://www.health.pa.gov/topics/documents/han/2018-pahan-400-03-02-adv_hiv%20sak.pdf).
33. New York State Department of Health. Health advisory: increased number of HIV diagnoses in Monroe County. Albany, NY: New York State Department of Health; October 15, 2020. [https://www.health.ny.gov/diseases/aids/providers/health\\_advisories/docs/health\\_advisory\\_monroe.pdf](https://www.health.ny.gov/diseases/aids/providers/health_advisories/docs/health_advisory_monroe.pdf).

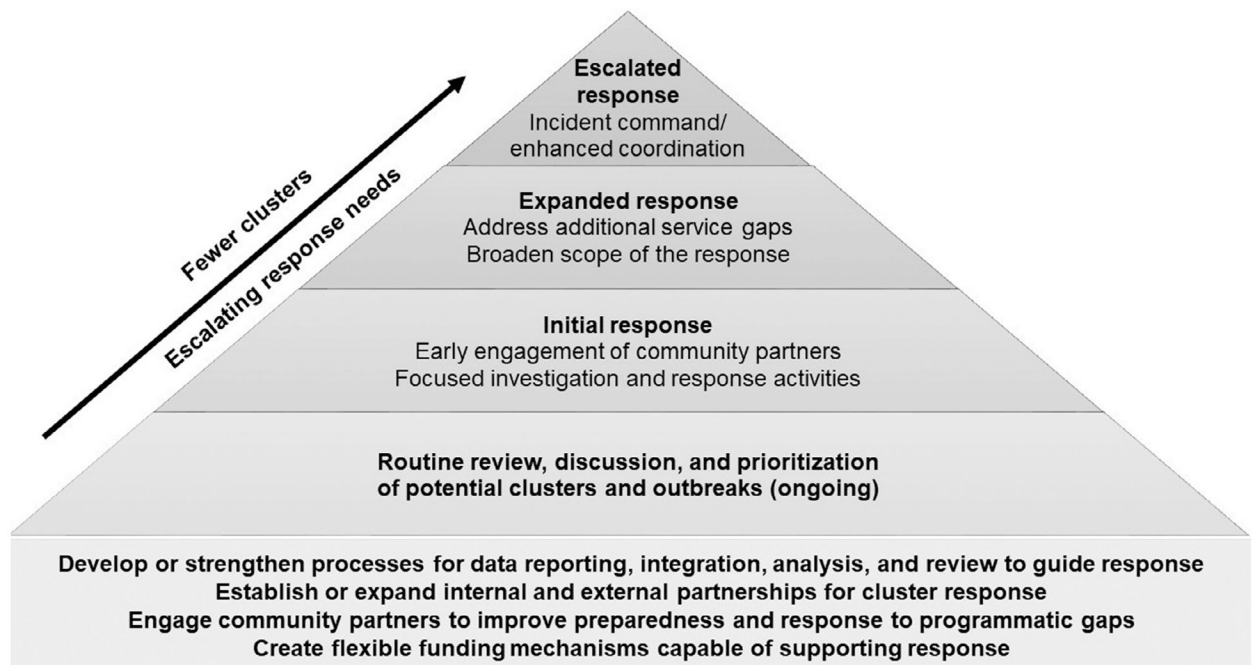
34. Dennis AM, Pasquale DK, Billock R, et al. Integration of contact tracing and phylogenetics in an investigation of acute HIV infection. *Sex Transm Dis*. 2018;45(4):222–228. 10.1097/OLQ.0000000000000726. [PubMed: 29465708]
35. Oster AM, France AM, Panneer N, et al. Identifying clusters of recent and rapid HIV transmission through analysis of molecular surveillance data. *J Acquir Immune Defic Syndr*. 2018;79(5):543–550. 10.1097/QAI.0000000000001856. [PubMed: 30222659]
36. Monterosso A, Minnerly S, Goings S, et al. Identifying and investigating a rapidly growing HIV transmission cluster in Texas. In: Paper presented at: Conference on Retroviruses and Opportunistic Infections; February 13–16; 2017. <https://www.croiconference.org/abstract/identifying-and-investigating-rapidly-growing-hiv-transmission-cluster-texas/>. Accessed August 10, 2021.
37. Poon AF, Gustafson R, Daly P, et al. Near real-time monitoring of HIV transmission hotspots from routine HIV genotyping: an implementation case study. *Lancet HIV*. 2016;3(5):e231–e238. 10.1016/S2352-3018(16)00046-1. [PubMed: 27126490]
38. Watson J, Brandt MG, Erskine J, Wilczynski C. Collaborative state and local health department response to intervene in a rapidly expanding molecular HIV cluster in Michigan. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 10, 2021.
39. Garcia M Project PODER: a Community-Based Approach for HIV Cluster Detection and Response Among Hispanic MSM and Transgender Communities Throughout Houston, TX. Paper presented at: AIDS 2020 –23rd International AIDS Conference; July 6–10; 2020. virtual <http://programme.aids2020.org/Abstract/Abstract/9505>. Accessed August 10, 2021.
40. Desai AN, Conyngham SC, Mashas A, et al. Interdisciplinary HIV sentinel case review: Identifying practices to prevent outbreaks in Philadelphia. *Am J Prev Med*. 2021;61(5S1):S151–S159. [PubMed: 34686284]
41. Flynn C, Pearlowitz ML, Stockdale J. Development and implementation of and findings from an integrated statewide HIV cluster detection and response system in Maryland. *Am J Prev Med*. In press.
42. Oster AM, France AM, Mermin J. Molecular epidemiology and the transformation of HIV prevention [published correction appears in *JAMA*. 2018;319(20):2138] *JAMA* 2018;319(16):1657–1658. 10.1001/jama.2018.1513. [PubMed: 29630701]
43. Ocfemia MCB, Cope S, Wertheim JO, Oster AM. Secure HIV-TRACE: a bioinformatics application for analyzing and visualizing molecular clusters of active HIV-1 transmission to guide prevention efforts. In: Paper presented at: 2018 Public Health Informatics Conference; August 22, 2018. <https://www.eventscribe.com/2018/Informatics/fsPopup.asp?efp=U0RXQIVPU1A2MTk5&PresentationID=418042&rnd=0.985011&mode=presinfo>. Accessed August 10, 2021.
44. Campbell E, Boyles A, Knyazev S, Shankar A, Switzer WM. Microbe-Trace: retooling molecular epidemiology for faster public health action. In: Paper presented at: 2019 National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
45. Tumpney M, John B, Panneer N, et al. Human immunodeficiency virus (HIV) outbreak investigation among persons who inject drugs in Massachusetts enhanced by HIV sequence data. *J Infect Dis*. 2020;222(suppl 5):S259–S267. 10.1093/infdis/jiaa053. [PubMed: 32877558]
46. Wertheim JO, Kosakovsky Pond SL, Forgione LA, et al. Social and genetic networks of HIV-1 transmission in New York City. *PLoS Pathog*. 2017;13(1):e1006000. 10.1371/journal.ppat.1006000. [PubMed: 28068413]
47. Pasquale DK, Doherty IA, Miller WC, et al. Factors associated with human immunodeficiency virus infections linked in genetic clusters but disconnected in partner tracing. *Sex Transm Dis*. 2020;47(2):80–87. 10.1097/OLQ.0000000000001094. [PubMed: 31934954]
48. Pasquale DK, Doherty IA, Sampson LA, et al. Leveraging phylogenetics to understand HIV transmission and partner notification networks. *J Acquir Immune Defic Syndr*. 2018;78(4):367–375. 10.1097/QAI.0000000000001695. [PubMed: 29940601]

49. Campbell EM, Jia H, Shankar A, et al. Detailed transmission network analysis of a large opiate-driven outbreak of HIV infection in the United States. *J Infect Dis.* 2017;216(9):1053–1062. 10.1093/infdis/jix307. [PubMed: 29029156]
50. France AM, Panneer N, Ocfemia MCB, et al. Rapidly growing HIV transmission clusters in the United States, 2013–2016. In: Paper presented at: 2020 Conference on Retroviruses and Opportunistic Infections; March 2018. <https://www.croiconference.org/abstract/rapidly-growing-hiv-transmission-clusters-united-states-2013-2016/>. Accessed August 10, 2021.
51. Erly SJ, Herbeck JT, Kerani RP, Reuer JR. Characterization of molecular cluster detection and evaluation of cluster investigation criteria using machine learning methods and statewide surveillance data in Washington state. *Viruses.* 2020;12(2):142. 10.3390/v12020142. [PubMed: 31991877]
52. McClung RP, Atkins AD, Kilkenny M, et al. Response to a large HIV outbreak, Cabell County, West Virginia, 2018–2019. *Am J Prev Med.* 2021;61(5S1):S143–S150. [PubMed: 34686283]
53. Dennis AM, Hué S, Billock R, et al. Human immunodeficiency virus type 1 phylodynamics to detect and characterize active transmission clusters in North Carolina. *J Infect Dis.* 2020;221(8):1321–1330. 10.1093/infdis/jiz176. [PubMed: 31028702]
54. Wertheim JO, Oster AM, Switzer WM, et al. Natural selection favoring more transmissible HIV detected in United States molecular transmission network. *Nat Commun.* 2019;10(1):5788. 10.1038/s41467-019-13723-z. [PubMed: 31857582]
55. Lyss SB, Buchacz K, McClung RP, Asher A, Oster AM. Responding to outbreaks of human immunodeficiency virus among persons who inject drugs—United States, 2016–2019: perspectives on recent experience and lessons learned. *J Infect Dis.* 2020;222(suppl 5):S239–S249. 10.1093/infdis/jiaa112. [PubMed: 32877545]
56. Recent HIV clusters and outbreaks across the United States among people who inject drugs and considerations during the COVID-19 pandemic. Centers for Disease Control and Prevention; March 27, 2020. <https://emergency.cdc.gov/han/2020/han00436.asp>.
57. Centers for Disease Control and Prevention. HIV Surveillance Report, 2019. <http://www.cdc.gov/hiv/library/reports/hiv-surveillance.html>. Published May 2021. Accessed August 10, 2021.
58. Hightow LB, MacDonald PD, Pilcher CD, et al. The unexpected movement of the HIV epidemic in the Southeastern United States: transmission among college students. *J Acquir Immune Defic Syndr.* 2005;38(5):531–537. 10.1097/01.qai.0000155037.10628.cb. [PubMed: 15793362]
59. Oster AM, Dorell CG, Mena LA, Thomas PE, Toledo CA, Heffelfinger JD. HIV risk among young African American men who have sex with men: a case-control study in Mississippi. *Am J Public Health.* 2011;101(1):137–143. 10.2105/AJPH.2009.185850. [PubMed: 21088266]
60. Des Jarlais DC, Sypsa V, Feelemyer J, et al. HIV outbreaks among people who inject drugs in Europe, North America, and Israel. *Lancet HIV.* 2020;7(6):e434–e442. 10.1016/S2352-3018(20)30082-5. [PubMed: 32504576]
61. Flynn C Investigating HIV sequence clusters across state lines; Maryland’s experience in regional cooperation with Virginia and the District of Columbia. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
62. Oster AM, France AM, McClung RP, et al. The CDC HIV outbreak coordination unit: developing a standardized, collaborative approach to HIV outbreak assessment and response. *Public Health Rep.* In press. Online May 28, 2021. 10.1177/00333549211018678.
63. Federal Emergency Management Authority. National incident management system. third edition Washington, DC: Federal Emergency Management Authority; October 10, 2017. [https://www.fema.gov/sites/default/files/2020-07/fema\\_nims\\_doctrine-2017.pdf](https://www.fema.gov/sites/default/files/2020-07/fema_nims_doctrine-2017.pdf).
64. Buskin SE, Erly SJ, Glick SN, et al. Detection and response to an HIV cluster: People living homeless and using drugs in Seattle, Washington. *Am J Prev Med.* 2021;61(5S1):S160–S169. [PubMed: 34686286]
65. Raney Z, Vogel S. The local perspective on the importance of communication during an HIV cluster investigation. In: Paper presented at: National HIV Prevention Conference; March 2019.

66. Carlson D, Raney Z, Grimes Trotter A, et al. Interstate response to a rapid increase of HIV among persons who inject drugs in northern Kentucky and Hamilton County, Ohio. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
67. Blau E Responding to increases in newly identified HIV infection among persons who inject drugs. In: Paper presented at: Association of Nurses in AIDS Care Conference 2019; November 2019.
68. McNeese M, Hallmark C, Liu Z, et al. Biomedical intervention meets community engagement: introducing molecular HIV surveillance (MHS) cluster analysis to community and stakeholders. In: Paper presented at: National HIV Prevention Conference; March 2019.
69. Watson J Shared HIV networks: leveraging Michigan's integrated Health Department to create an innovative molecular HIV surveillance system. Paper presented at: SYNChronicity Conference; October 2020. <https://sync2020.org/wp-content/uploads/2020/10/Innovative-Practice-Session-3-Watson.pdf>.
70. Petran E, Rybicki S. Addressing the HIV outbreak among people with HIV and the American Indian community. ATEC AIDS Education & Training Center Program, National Coordinating Resource Center; November 17, 2020. <https://aidsetc.org/blog/addressing-hiv-outbreak-among-people-hiv-and-american-indian-community-matec-minnesotas-hiv>.
71. Duwve J, Stover D. HIV outbreak related to injection of prescription opioids in Scott County, Indiana: an updated emphasizing the power of partnerships. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
72. Pack A, Venegas Y, Ortega J, Taylor B, Woo C. Community response to HIV clusters in San Antonio, Texas, 2017–2018. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
73. Mohammad Khan I, Skinner DS, Cote CG, et al. Mobilizing a Multijurisdictional Response to Address Alarming Rates of HIV and Hepatitis C (HCV) in Rural Saskatchewan (SK) Indigenous Community (IC) in Canada. In: Paper presented at: AIDS 2018; 2018. July 23–27.
74. Wertheim JO, Murrell B, Mehta SR, et al. Growth of HIV-1 molecular transmission clusters in New York City. *J Infect Dis.* 2018;218 (12):1943–1953. 10.1093/infdis/jiy431. [PubMed: 30010850]
75. Dennis AM, Frost SDW, Cressman A, et al. Statewide HIV-1 transmission cluster detection and prioritization for response. In: Paper presented at: Conference on Retroviruses and Opportunistic Infections; 2019. March 4–9 [https://2jg4quetidw2blbbq2ixwziw-wpengine.netdna-ssl.com/wp-content/uploads/sites/2/posters/2020/1430\\_2\\_Dennis\\_00908.pdf](https://2jg4quetidw2blbbq2ixwziw-wpengine.netdna-ssl.com/wp-content/uploads/sites/2/posters/2020/1430_2_Dennis_00908.pdf).
76. Hunsucker D Injection drug use, hepatitis, and HIV: preparing for and addressing vulnerability in a rural community. In: Paper presented at: National HIV Prevention Conference; 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
77. National Association of County and City Health Officials. Community response planning for outbreaks of hepatitis and HIV among people who inject drugs: a case study from LENOWISCO Health District, a rural community in Virginia. Washington, DC: National Association of County and City Health Officials; 2018. [https://www.naccho.org/uploads/downloadable-resources/LENOWISCO-Project-Report\\_2018\\_with-appendices.pdf](https://www.naccho.org/uploads/downloadable-resources/LENOWISCO-Project-Report_2018_with-appendices.pdf).
78. Dailey A, Richman R, Lane L. Using HIV surveillance data to operationalize a cluster response to detect and interrupt HIV transmission among persons who inject drugs. In: Paper presented at: National HIV Prevention Conference; March 2019. <https://www.cdc.gov/nhpc/pdf/NHPC-2019-Abstract-Book.pdf>. Accessed August 13, 2021.
79. Curran KG, Saduvala N, Linley L, Peruski A, France AM, Oster AM. Real-time HIV case and sequence reporting for cluster detection: United States. In: Paper presented at: Conference on Retroviruses and Opportunistic Infections; March 2021. <https://www.croiconference.org/abstract/timely-hiv-case-and-sequence-reporting-for-cluster-detection-united-states/>. Accessed August 10, 2021.
80. Quilter LAS, Agnew-Brune C, Broussard D, et al. Establishing best practices in a response to an HIV cluster: an example from a surge response in West Virginia. *Sex Transm Dis.* 2021;48(3):e35–e40. 10.1097/OLQ.0000000000001279. [PubMed: 32890333]

81. Oregon Health Authority. HIV in people who inject drugs. Salem, OR: Oregon Health Authority; August 29, 2019. <https://www.oregon.gov/oha/ph/diseasesconditions/communicabledisease/diseasesurveillancedata/hivdata/documents/hivslides.pdf>.
82. France AM, Gopalappa C, Singh S, et al. Nationally Representative Simulation Model of HIV Transmission Clusters in the United States. Paper presented at: AIDS 2020. [https://www.aids2020.org/wp-content/uploads/2020/09/AIDS2020\\_Abstracts.pdf](https://www.aids2020.org/wp-content/uploads/2020/09/AIDS2020_Abstracts.pdf). Accessed August 10, 2021
83. Wertheim JO, Panneer N, France AM, Saduvala N, Oster AM. Incident infection in high-priority HIV molecular transmission clusters in the United States. *AIDS*. 2020;34(8):1187–1193. 10.1097/QAD.0000000000002531. [PubMed: 32287065]
84. Kimbrough LW, Fisher HE, Jones KT, Johnson W, Thadiparthi S, Dooley S. Accessing social networks with high rates of undiagnosed HIV infection: the social networks demonstration project. *Am J Public Health*. 2009;99(6):1093–1099. 10.2105/AJPH.2008.139329. [PubMed: 19372521]
85. Wohl AR, Ludwig-Barron N, Dierst-Davies R, et al. Project Engage: snowball sampling and direct recruitment to identify and link hard-to-reach HIV-infected persons who are out of care. *J Acquir Immune Defic Syndr*. 2017;75(2):190–197. 10.1097/QAI.0000000000001312. [PubMed: 28169872]
86. Bradley H, Hogan V, Agnew-Brune C, et al. Increased HIV diagnoses in West Virginia counties highly vulnerable to rapid HIV dissemination through injection drug use: a cautionary tale. *Ann Epidemiol*. 2019;34:12–17. 10.1016/j.annepidem.2019.02.012. [PubMed: 30967302]
87. Broz D, Zibbell J, Foote C, et al. Multiple injections per injection episode: high-risk injection practice among people who injected pills during the 2015 HIV outbreak in Indiana. *Int J Drug Policy*. 2018;52:97–101. 10.1016/j.drugpo.2017.12.003. [PubMed: 29278838]
88. Balaji AB, Oster AM, Viall AH, Heffelfinger JD, Mena LA, Toledo CA. Role flexing: how community, religion, and family shape the experiences of young black men who have sex with men. *AIDS Patient Care STDS*. 2012;26(12):730–737. 10.1089/apc.2012.0177. [PubMed: 23134543]
89. Dorell CG, Sutton MY, Oster AM, et al. Missed opportunities for HIV testing in health care settings among young African American men who have sex with men: implications for the HIV epidemic. *AIDS Patient Care STDS*. 2011;25(11):657–664. 10.1089/apc.2011.0203. [PubMed: 21923415]
90. Oster AM, Wejnert C, Mena LA, Elmore K, Fisher H, Heffelfinger JD. Network analysis among HIV-infected young black men who have sex with men demonstrates high connectedness around few venues. *Sex Transm Dis*. 2013;40(3):206–212. 10.1097/OLQ.0b013e3182840373. [PubMed: 23403601]
91. Little SJ, Kosakovsky Pond SL, Anderson CM, et al. Using HIV networks to inform real time prevention interventions. *PLoS One*. 2014;9(6):e98443. 10.1371/journal.pone.0098443. [PubMed: 24901437]
92. Harbarger M Spike in Multnomah County HIV cases tied to drug use. *OregonLive*. June 20, 2019 <https://www.oregonlive.com/health/2019/06/spike-in-multnomah-county-hiv-cases-tied-to-drug-use.html>.
93. Buskin S, Lechtenberg R, Golden MR, Glick SN. Increase in HIV among people who inject drugs in 2018 and 2019: cluster identification and response. <https://www.kingcounty.gov/depts/health/communicable-diseases/hiv-std/patients/~media/depts/health/communicable-diseases/documents/hivstd/2018-hiv-aids-epidemiology-annual-report.ashx>. Accessed January 16, 2021.
94. Melvin SC, Gipson J. The Open Arms Healthcare Center's integrated HIV care services model. *Prev Chronic Dis*. 2019;16:E135. 10.5888/pcd16.180633. [PubMed: 31580796]
95. Furukawa NW, Weimer M, Willenburg KS, et al. Expansion of pre-exposure prophylaxis capacity in response to an HIV outbreak among people who inject drugs-Cabell County, West Virginia, 2019. *Public Health Rep*. March 1, 2021 In press. Online. 10.1177/0033354921994202.
96. Menza TW. HIV and other infectious diseases among people who use drugs. Salem, OR: Oregon Health Authority; July 3, 2019. <https://www.oregon.gov/oha/ph/diseasesconditions/communicabledisease/diseasesurveillancedata/hivdata/documents/pcPLetter.pdf>.

97. HIV investigation news releases. Northern Kentucky Health Department. <https://nkyhealth.org/individual-or-family/health-alerts/hiv-investigation/>. Updated January 22, 2019. Accessed January 16, 2021.
98. Dasgupta S, Broz D, Tanner M, et al. Changes in reported injection behaviors following the public health response to an HIV outbreak among people who inject drugs: Indiana, 2016. *AIDS Behav*. 2019;23(12):3257–3266. 10.1007/s10461-019-02600-x. [PubMed: 31313095]
99. Patel MR, Foote C, Duwve J, et al. Reduction of injection-related risk behaviors after emergency implementation of a syringe services program during an HIV outbreak. *J Acquir Immune Defic Syndr*. 2018;77(4):373–382. 10.1097/QAI.0000000000001615. [PubMed: 29271829]
100. Watson M, Sweeney P. Furthering discussion of ethical implementation of HIV cluster detection and response. *Am J Bioeth*. 2020;20 (10):24–26. 10.1080/15265161.2020.1806398.
101. Mermin J, Valentine SS, McCray E. HIV criminalisation laws and ending the US HIV epidemic. *Lancet HIV*. 2021;8(1):e4–e6. 10.1016/S2352-3018(20)30333-7. [PubMed: 33387478]
102. National Association of State and Territorial AIDS Directors. HIV data privacy and confidentiality: legal & ethical considerations for health department data sharing. Washington, DC: National Association of State and Territorial AIDS Directors; June 2018. <https://www.nastad.org/sites/default/files/Uploads/2018/nastad-hiv-data-privacy-06062018.pdf>.



**Figure 1.**  
Scope of HIV cluster detection and response activities.

**Table 1.** HIV Cluster and Outbreak Response Interventions to Understand Transmission Networks: Key Examples

| Response interventions         | Potential collaborators                                                                                                                                 | Potential approaches                                                                                                                                          | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Review of existing HD data     | <ul style="list-style-type: none"> <li>Partner services, STD, viral hepatitis, or injury programs</li> <li>HIV prevention or care program</li> </ul>    | <ul style="list-style-type: none"> <li>Analysis of partner services data</li> <li>Analysis of HIV, STD, viral hepatitis, or drug overdose data</li> </ul>     | <p>Key example: in a response to a molecular cluster of 27 MSM in San Antonio, Texas, analysis of existing partner services data revealed that the network included an additional 88 people with HIV, and many people without HIV but at risk for HIV were likely in the underlying risk network.<sup>36</sup></p> <p>Other examples: network analysis and visualization of molecular and partner services data in Tennessee and North Carolina to assess transmission networks.<sup>19,34</sup> Use of diverse sources of public health data to understand social and structural issues contributing to a Philadelphia outbreak.<sup>30</sup></p>                                                                                                                                                                                                                                                                                                                                          |
| HIV partner services expansion | <ul style="list-style-type: none"> <li>STD program</li> <li>Local HDs</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>Surge capacity</li> <li>Reinterview/enhanced interview</li> </ul>                                                      | <p>Key example: A 2017 response among MSM in West Virginia featured intensified partner services, including (1) eliciting an expanded network of contacts, (2) reinterviewing previously interviewed persons, and (3) testing and interviewing contacts of people in the network who did not have HIV. This revealed potential for onward IDU-related transmission through connections with PWID.<sup>81,87</sup></p> <p>Other examples: surge capacity to reach hundreds of contacts in Indiana and West Virginia (2019) responses.<sup>11,52</sup> Partner services reinterview yielding additional contacts in British Columbia.<sup>37</sup></p> <p>Outreach to providers and cluster members to expand partner elicitation in Michigan.<sup>38</sup> Reinterviewing partners who had previously tested negative in San Antonio.<sup>36</sup> Embedding partner services staff in an SSP during a response in Miami.<sup>17</sup></p>                                                   |
| Peer-recruitment strategies    | <ul style="list-style-type: none"> <li>Local HDs</li> <li>Community-based organizations</li> <li>Academic institutions</li> </ul>                       | <ul style="list-style-type: none"> <li>Social network strategy</li> </ul>                                                                                     | <p>Key example: A 2019 response among PWID in West Virginia included a peer-recruitment program using coupons and food-based incentives to recruit PWID to the SSP for enrollment, HIV testing, and PrEP education and linkage.<sup>52</sup></p> <p>Other examples: social network strategy to recruit Hispanic/Latino MSM in clusters for services in Houston, Texas.<sup>39</sup></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Rapid qualitative assessment   | <ul style="list-style-type: none"> <li>Community-based organizations</li> <li>Local HDs</li> <li>Academic institutions</li> </ul>                       | <ul style="list-style-type: none"> <li>Qualitative interviews with the affected population</li> <li>Qualitative interviews with other stakeholders</li> </ul> | <p>Key example: in Massachusetts, qualitative interviews with PWID, providers, and other stakeholders revealed information about changes in drug supply that had led to increased injection frequency and likely facilitated transmission; these interviews also identified challenges in HIV treatment services and SSP accessibility that were addressed during the response.<sup>15,16</sup></p> <p>Other examples: qualitative interviews shedding light on emerging drug behaviors in Indiana.<sup>88</sup> Rapid needs assessment leading to expansion of medical and social services in Seattle.<sup>65</sup></p> <p>Qualitative interviews highlighting stigma, discrimination, and homophobia in Mississippi and Philadelphia.<sup>40,89</sup> Qualitative interviews revealing unstable housing and a lack of knowledge about prevention services among PWID and gaps in understanding regarding SSPs among the general community in northern Kentucky and Ohio.<sup>68</sup></p> |
| Quantitative data collection   | <ul style="list-style-type: none"> <li>Healthcare facilities/providers</li> <li>Community-based organizations</li> <li>Academic institutions</li> </ul> | <ul style="list-style-type: none"> <li>Medical record review</li> <li>Supplemental questionnaires</li> </ul>                                                  | <p>Key example: medical record review in northern Kentucky and Ohio identified missed opportunities for HIV testing during clinical encounters, leading to expansion of HIV testing of PWID in emergency departments.<sup>22</sup></p> <p>Other examples: medical record review identifying gaps in PrEP in San Antonio<sup>36</sup> and experiences with HIV, STI, and mental-health services in Philadelphia.<sup>40</sup> Collection and analysis of survey and network data in Tennessee, Mississippi, and North Carolina.<sup>5,19,59,60,90,91</sup></p>                                                                                                                                                                                                                                                                                                                                                                                                                               |

HD, health department; IDU, injection drug use; MSM, men who have sex with men; PrEP, pre-exposure prophylaxis; PWID, people who inject drugs; SSP, syringe services program; STD, sexually transmitted disease; STI, sexually transmitted infection.

**Table 2.** HIV Cluster and Outbreak Response Interventions to Address Gaps in Programs and Services: Key Examples

| Response interventions               | Potential collaborators                                                                                                                                  | Potential approaches                                                                                                                                                                                                                                | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HIV testing expansion                | <ul style="list-style-type: none"> <li>Healthcare facilities and providers</li> <li>Public health laboratory</li> <li>CBOs</li> <li>Local HDs</li> </ul> | <ul style="list-style-type: none"> <li>Testing in EDs, clinics, and correctional facilities</li> <li>Venue-based or other outreach, including mobile units</li> </ul>                                                                               | <p>Key example: Massachusetts doubled the number of field epidemiologists, expanded partner services to all new diagnoses, and implemented HIV testing in EDs, homeless shelters, and jails.<sup>16</sup></p> <p>Other examples: establishing new testing sites in Indiana.<sup>72</sup> Expansion of HIV testing of PWID in EDs in Northern Kentucky.<sup>68</sup> HIV testing expansion to multiple locations in West Virginia.<sup>52</sup> HIV testing expansion in EDs, jails, SSPs, homeless encampments, social-service organizations, and a mobile clinic in Seattle.<sup>16,65</sup> Increased community testing at mobile units and fixed sites in Philadelphia.<sup>30</sup> Expanded testing at SSPs in North Carolina.<sup>23</sup> Opt-out testing at county safety-net hospital in San Antonio.<sup>73</sup> Outreach to homeless encampments for mobile testing in Multnomah County, Oregon.<sup>93</sup></p> |
| HIV and other medical care expansion | <ul style="list-style-type: none"> <li>Healthcare facilities/providers</li> <li>Ryan White HIV/AIDS Program</li> </ul>                                   | <ul style="list-style-type: none"> <li>Low-barrier care</li> <li>Care navigation</li> <li>Care coordination</li> </ul>                                                                                                                              | <p>Key example: in Miami, the response team implemented onsite services to improve linkage to HIV care and viral suppression. Flexibility and continuous quality improvement led to the design of a culturally appropriate navigation program for a population experiencing challenges in accessing care in a complex healthcare system.<sup>17</sup></p> <p>Other examples: first-time initiation of HIV medical care in Scott County, Indiana.<sup>72</sup> Identification of one-stop shops providing comprehensive services, including HIV treatment and PrEP, in Philadelphia.<sup>30</sup> Addition of low-barrier HIV care through a mobile clinic for women experiencing homelessness, substance use disorders, or exchange sex in Seattle.<sup>94</sup> Enhanced care coordination in West Virginia leading to a personalized, multidisciplinary care system.<sup>52</sup></p>                                       |
| PrEP expansion                       | <ul style="list-style-type: none"> <li>Surge capacity</li> <li>Healthcare facilities/providers</li> <li>CBOs</li> </ul>                                  | <ul style="list-style-type: none"> <li>New PrEP sites</li> <li>Provider PrEP training</li> <li>Targeted PrEP campaigns</li> </ul>                                                                                                                   | <p>Key example: in a 2019 West Virginia response among PWID, PrEP access was expanded through training &gt;100 providers and supporting PrEP implementation.<sup>52,96</sup></p> <p>Other examples: increase in the number of PrEP sites from 4 to 17 in San Antonio.<sup>73</sup> Expansion of PrEP provision at STD clinics, SSPs, and a mobile clinic in Seattle.<sup>94</sup></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| SSP expansion                        | <ul style="list-style-type: none"> <li>HD or non-HD SSP programs</li> </ul>                                                                              | <ul style="list-style-type: none"> <li>Expanded options (fixed site, mobile van, delivery, secondary distribution)</li> <li>Policy or law change to allow or ease approval of SSPs</li> <li>Expanded hours</li> <li>Needs-based exchange</li> </ul> | <p>Key example: in an Indiana response, laws were modified to allow the first SSP in the state, which provided treatment for opioid use disorder, hepatitis A vaccination, linkage to HIV and hepatitis C virus testing and treatment, and naloxone distribution. The SSP contributed to substantial decreases in unsafe injection practices, reduced HIV transmission, prevented hepatitis A despite an outbreak in neighboring counties, and reduced endocarditis admissions.<sup>72,99,100</sup></p> <p>Other examples: change in state law leading to expansion of SSPs across the state in Massachusetts.<sup>16,55</sup> Changes in SSP policies to expand eligibility and increase syringes provided in West Virginia.<sup>52</sup> Establishment of a new mobile SSP in Seattle.<sup>94</sup> Increased funding for local SSP and doubling of capacity for exchange in Philadelphia.<sup>30</sup></p>                 |
| Other interventions                  | <ul style="list-style-type: none"> <li>Social services providers</li> <li>Medicaid program</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Expansion of health insurance</li> <li>Expansion of social support services</li> </ul>                                                                                                                       | <p>Key example: Indiana's multitiered response included immediate access to health insurance, addressing an important structural barrier to care and prevention.<sup>72</sup></p> <p>Other examples: expanded services in Massachusetts that address social instability related to homelessness and incarceration.<sup>15</sup> Establishment of a new outreach center providing food, transportation, family, referral, and cultural support in Saskatchewan.<sup>74</sup> Training &gt;100 clinicians and educating lawmakers on LGBT and sexual health and founding the first LGBT health clinic in Mississippi's history.<sup>89,95</sup></p>                                                                                                                                                                                                                                                                             |



**Table 3.**  
Research Gaps Related to Cluster Detection and Response

| Research gaps                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identify new or improved methods for timely cluster detection using new or existing data sources and combinations of existing approaches (e.g., molecular, time-space, partner services).                                                                                                                                                                                                                                               |
| Describe the contribution of HIV clusters and outbreaks to overall HIV incidence in the U.S. and further characterize the factors facilitating rapid HIV transmission in clusters and outbreaks.                                                                                                                                                                                                                                        |
| Use simulation, modeling, or other analytic methods to predict the future occurrence and trajectory of clusters and the relative impact of specific HIV prevention and care services, policies, and external factors on preventing or mitigating clusters and outbreaks.                                                                                                                                                                |
| Develop deployable interventions to rapidly address specific gaps during cluster/outbreak response (e.g., HIV testing, PrEP, syringe services, rapid linkage, and treatment initiation) and assess the relative impact on HIV transmission through modeling and other analyses. Interventions could include peer-driven recruitment strategies, standardized tools for data collection, and improved partner services outreach methods. |
| Assess the direct impact of cluster detection and response programs on interrupting or preventing HIV transmission, on specific HIV prevention and care outcomes (e.g., PrEP use, viral suppression), and on outcomes for vulnerable populations. Assess common programmatic gaps identified during responses and changes to programs, policies, or procedures resulting from cluster response.                                         |
| Develop new or improved approaches to evaluating outcomes during and after cluster response, including developing specific indicators or metrics for real-time monitoring during a response and evaluation after a response, and cost/cost-effectiveness impact on HIV programs.                                                                                                                                                        |
| Identify effective models for engaging the public, key stakeholders, and specific communities in HIV outbreak preparedness, planning, and response, including approaches that identify and address HIV stigma, structural barriers, and social determinants of health; community-based participatory models; and improved models for collaboration with healthcare systems and providers and nonpublic health stakeholders.             |
| Identify effective models for oversight, management, and implementation of cluster detection and response programs, including models for HIV emergency response, staff surge, multijurisdictional HIV outbreak response, and data management and data sharing during a response.                                                                                                                                                        |

PrEP, pre-exposure prophylaxis.